Satyam Jain

AI Engineer · Data Scientist · Agentic AI & LLM Safety

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SUMMARY

AI engineer and data scientist with 4+ years shipping production generative AI. I build **agentic AI systems** — multi-agent orchestration, tool-calling agents, RAG and human-in-the-loop workflows — and the **guardrails** that make them safe to deploy: content-safety classifiers, prompt-injection defense, PII detection and policy enforcement for agent traffic. M.Tech in Data Science, with hands-on work in model evaluation, class-imbalanced learning and fine-tuning on datasets up to 3.9M samples. Author of **pii-protector** (PyPI) and 35 open models on Hugging Face.

CORE COMPETENCIES

AGENTIC AI	AI agents, multi-agent systems, agent orchestration, LangGraph, LangChain, tool / function calling, MCP (Model Context Protocol), agent routing, stateful workflows, agent memory, human-in-the-loop, agent SDKs
AGENT SAFETY	Guardrails, prompt-injection & jailbreak defense, tool & MCP allowlists, action gating, access control for retrieval, PII redaction, red teaming, audit trails
GENAI & LLMS	RAG, vector search, embeddings, Claude, GPT, Llama, Qwen, Gemma, Amazon Bedrock, vLLM, Ollama, prompt engineering, LLM evaluation
DATA SCIENCE & ML	Fine-tuning (LoRA, QLoRA), multi-task learning, class imbalance, focal loss, model benchmarking, error analysis, NLP, classification, anomaly detection, EDA, statistics

EXPERIENCE

AI Engineer — SAF3AI Inc, Indore

Jan 2026 – Present

- Designed **multi-agent orchestration** systems with task delegation and inter-agent communication; built reusable **agent SDKs** that cut prototyping time by ~40%.
- Contributed to an **AI gateway** that governs LLM and agent traffic: provider and **tool allowlists**, **MCP-server allowlists**, request / token / cost limits, and per-request audit events.
- Built **content-safety guardrails** for text and image, running inline in production with per-tenant policy thresholds and audit logging.
- Implemented LLM security-testing frameworks for prompt injection, data leakage and model-behavior analysis; built compliance tooling with automated policy checks.

Senior Gen AI Developer — NOBORDER.z Innovations, Indore

Jul 2022 – Nov 2025

- Led **AvatarChat**, a multilingual customer-support **AI agent**: RAG over company documentation, **tool-driven ticket creation**, human handoff on low confidence, and TTS with a 2D avatar.
- Architected a **multi-tenant chatbot platform on Amazon Bedrock** (Claude + Titan) with per-tenant isolation, vector RAG and a fine-tuning pipeline, serving **1,000+ concurrent users**.
- Built an automated interview platform: ATS resume screening and scoring, dynamic assessment generation, real-time LLM answer evaluation and video-based proctoring.
- Delivered voice AI for Xana Genesis (Japanese TTS via VoiceVox, multilingual STT with noise removal) on AWS EC2 with Docker; built 3D-asset retrieval with sentence-transformer similarity.

Python Developer — Webllisto Technologies, Indore

Jan 2022 – Mar 2023

- Developed Python modules for an ERP system, improving operational efficiency by 30%; built a Django e-commerce platform with auth, payments and inventory.

AI AGENT PROJECTS

Context Layer for Enterprise Agents

github.com/sjain26/ninebar-context-layer

Context layer that decides what an agent may see and do. Every fact carries its source; five gates run in order (**access policy → injection scan → staleness → conflict resolution → action gate**). Retrieved text never enters the instruction channel, and actions are gated by reversibility and blast radius (auto / confirm / approve / block), not model confidence. 25 tests and 15 scenario evals.

Multi-Agent Financial Research System

finance-agent-chapter247.streamlit.app

LangGraph pipeline of 6 specialised agents (query analysis, web search, stock data, vector search, analysis, report) sharing typed state, with conditional routing and fallbacks at every stage. Tools: Alpha Vantage, DuckDuckGo, Pinecone.

AI Accounting Migration System github.com/sjain26/accounting-migration-system

Deterministic rules, dual-model LLM inference and **semantic memory**, with **human-in-the-loop** review. Auto-approves 70–80% of chart-of-accounts mappings after 3 runs on similar clients and flags 10+ journal anomaly types.

Multi-Agent Ordering Chatbot huggingface.co/spaces/jainsatyam26/intent-identifier

Orchestrator agent classifies intent and routes to specialised order agents (LangChain + Groq), with SQLite persistence, input sanitisation, rate limiting and session isolation.

DATA SCIENCE & RESEARCH

Generalisation Gap in Prompt-Injection Detectors HF dataset: prompt-injection-comparison

Reproducible benchmark harness across a 5,000-sample public set and a 500-sample custom attack set. Showed the top model's F1 dropping from 97.6 to 41.8 (recall 26.4%) on unseen attack styles, a **distribution-shift** failure that leaderboard scores hide.

State-Space vs. Transformer Under Class Imbalance HF: mamba-shield-v2, xlmr-shield

Dual-head (binary + 11-class multi-label) classifiers on 285K samples with resampling, per-class caps and weighted sampling. A **9.6M-parameter Mamba-2** reached 97% of a 278M XLM-R's safety accuracy (83.1% vs 85.6%) with 29× fewer parameters.

Large-Scale Multi-Task & PEFT Classifiers HF: guard-safety-classifier, sentinel-v2, gemma4-guardrail-v5

DeBERTa-v3 with 3 jointly trained heads on **3.9M samples** (multi-label F1 0.907, 92.8% binary accuracy). Qwen3-0.6B EN/HI/Hinglish injection detector at 0.90 average F1 and 48 ms latency. Gemma-4 **LoRA** fine-tune with focal loss on 151K samples.

SMS Spam Detection & Healthcare Text-to-SQL Analytics github.com/sjain26

Benchmarked 11 classifiers on TF-IDF features; a soft-voting ensemble reached **98.2% accuracy / 99.2% precision**, served via FastAPI. Built a natural-language analytics app that generates SQL / pandas queries over joined healthcare datasets, with an evaluation suite for SQL correctness.

OPEN SOURCE

pii-protector — PyPI, v2.2.5 pypi.org/project/pii-protector

PII detection with a 4-layer sequential ensemble: 90%+ of text resolves in a sub-3 ms regex layer, escalating to Presidio/spaCy and transformer NER only when needed. 50+ PII types including Aadhaar, PAN, GSTIN, cards and API secrets.

35 open models & 13 datasets on Hugging Face huggingface.co/jainsatyam26

Guardrail, safety and prompt-injection classifiers (Qwen, Gemma, Mistral, Llama Guard, DeBERTa, XLM-R, Mamba), including INT8 / ONNX builds for low-latency serving.

TECHNICAL SKILLS

LANGUAGES	Python, SQL, JavaScript, C++
AGENTS & LLM	LangGraph, LangChain, OpenAI SDK, Claude SDK, Groq, tool calling, MCP, RAG, Pinecone, Qdrant, embeddings, n8n
ML & DL	PyTorch, Transformers, PEFT (LoRA / QLoRA), TRL, scikit-learn, XGBoost, TensorFlow, Keras, Mamba / SSMs
DATA	pandas, NumPy, Matplotlib, Seaborn, PostgreSQL, SQLite, Jupyter, Streamlit
AI SAFETY	Llama Guard, Prompt Guard, Presidio, spaCy, NLTK, guardrail fine-tuning, red teaming
SERVING & CLOUD	vLLM, Ollama, ONNX Runtime, INT8 quantization, FastAPI, Docker, AWS (Bedrock, EC2, SageMaker, Transcribe)
VOICE & VISION	Whisper, VoiceVox, TTS / STT, OpenCV, Stable Diffusion, ControlNet

EDUCATION

M.Tech, Data Science — School of Data Science & Forecasting, DAVV, Indore

B.E., Electronics & Communication — Shri Vaishnav Institute of Technology & Science, Indore 2013 – 2016